WEBTECH LAB-6

1.

CODE:

1.test> use vehicles

```
test> use vehicles
switched to db vehicles
```

2. vehicles> show dbs

```
admin   40.00 KiB
config  60.00 KiB
local   40.00 KiB
```

3. vehicles> db.createCollection("two_wheelers", { capped: true, size: 50000, mmax: 100 })

{ ok: 1 }

vehicles> db.createCollection("four_wheelers")

{ ok: 1 }

4. vehicles> db.two_wheelers.insertMany([

| {

| bike_name: "Activa 6G",

| model: "gearless",

| category: "125cc",

| colors_available: ["red","black","blue"],

| manufacturer: "Honda",

| performance: 8,

| timestamp: new Date("2020-01-15"),

| price: 85000

| },

|

| {

| bike_name: "Pulsar 150",

| model: "gear",

| category: "150cc",

| colors_available: ["black","blue"],

| manufacturer: "Bajaj",

| performance: 9,

| timestamp: new Date("2019-05-10"),

| price: 110000

| },

|

| {

| bike_name: "Apache RTR 160",

| model: "gear",

| category: "150cc",

| colors_available: ["red","black"],

| manufacturer: "TVS",

| performance: 9,

| timestamp: new Date("2021-03-20"),

| price: 120000

| },

|

| {

| bike_name: "Jupiter",

| model: "gearless",

```
| category: "125cc",
| colors_available: ["blue","white"],
| manufacturer: "TVS",
| performance: 8,
| timestamp: new Date("2018-07-11"),
| price: 78000
| },
|
| {
| bike_name: "Splendor Plus",
| model: "gear",
| category: "100cc",
| colors_available: ["black","red"],
| manufacturer: "Hero",
| performance: 7,
| timestamp: new Date("2017-02-01"),
| price: 72000
| }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b19624072b2d5a197c2907'),
    '1': ObjectId('69b19624072b2d5a197c2908'),
    '2': ObjectId('69b19624072b2d5a197c2909'),
    '3': ObjectId('69b19624072b2d5a197c290a'),
    '4': ObjectId('69b19624072b2d5a197c290b')
```

```
  }
}
5. vehicles> db.four_wheelers.insertMany([
| {
| vehicle_name: "Swift",
| model: "own",
| category: "car",
| variants: ["vxi","zxi","petrol"],
| manufacturer: "Maruti",
| performance: 8,
| timestamp: new Date("2021-06-10"),
| price: 750000
| },
|
| {
| vehicle_name: "Innova",
| model: "own",
| category: "car",
| variants: ["diesel","zxi"],
| manufacturer: "Toyota",
| performance: 9,
| timestamp: new Date("2020-08-15"),
| price: 1800000
| },
|
| {
| vehicle_name: "Tata Bus",
```

| model: "commercial",
| category: "bus",
| variants: ["diesel"],
| manufacturer: "Tata",
| performance: 7,
| timestamp: new Date("2018-09-01"),
| price: 3500000
| },
|
| {
| vehicle_name: "Eicher Truck",
| model: "commercial",
| category: "heavy truck",
| variants: ["diesel"],
| manufacturer: "Eicher",
| performance: 8,
| timestamp: new Date("2019-11-05"),
| price: 2800000
| },
|
| {
| vehicle_name: "Bolero Pickup",
| model: "commercial",
| category: "mini truck",
| variants: ["diesel"],
| manufacturer: "Mahindra",
| performance: 8,

| timestamp: new Date("2022-01-10"),

| price: 900000

| }

|])

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b1962d072b2d5a197c290c'),
    '1': ObjectId('69b1962d072b2d5a197c290d'),
    '2': ObjectId('69b1962d072b2d5a197c290e'),
    '3': ObjectId('69b1962d072b2d5a197c290f'),
    '4': ObjectId('69b1962d072b2d5a197c2910')
  }
}
```

6.

vehicles> db.two_wheelers.find()

| db.four_wheelers.find()

```
{
  _id: ObjectId('69b1962d072b2d5a197c290c'),
  vehicle_name: 'Swift',
  model: 'own',
  category: 'car',
  variants: [ 'vxi', 'zxi', 'petrol' ],
  manufacturer: 'Maruti',
  performance: 8,
  timestamp: ISODate('2021-06-10T00:00:00.000Z'),
  price: 750000
},
{
  _id: ObjectId('69b1962d072b2d5a197c290d'),
  vehicle_name: 'Innova',
  model: 'own',
  category: 'car',
  variants: [ 'diesel', 'zxi' ],
  manufacturer: 'Toyota',
  performance: 9,
  timestamp: ISODate('2020-08-15T00:00:00.000Z'),
  price: 1800000
},
{
  _id: ObjectId('69b1962d072b2d5a197c290e'),
  vehicle_name: 'Tata Bus',
  model: 'commercial',
  category: 'bus',
  variants: [ 'diesel' ],
  manufacturer: 'Tata',
  performance: 7,
  timestamp: ISODate('2018-09-01T00:00:00.000Z'),
},
{
  _id: ObjectId('69b1962d072b2d5a197c290f'),
  vehicle_name: 'Eicher Truck',
  model: 'commercial',
  category: 'heavy truck',
  variants: [ 'diesel' ],
  manufacturer: 'Eicher',
  performance: 8,
  timestamp: ISODate('2019-11-05T00:00:00.000Z'),
  price: 2800000
},
{
  _id: ObjectId('69b1962d072b2d5a197c2910'),
  vehicle_name: 'Bolero Pickup',
  model: 'commercial',
  category: 'mini truck',
  variants: [ 'diesel' ],
  manufacturer: 'Mahindra',
  performance: 8,
  timestamp: ISODate('2022-01-10T00:00:00.000Z'),
  price: 900000
}
]
```

7. vehicles> db.two_wheelers.find({}, {bike_name:1, price:1, _id:0})

|

| db.four_wheelers.find({}, {vehicle_name:1, price:1, _id:0})

```
[
  { vehicle_name: 'Swift', price: 750000 },
  { vehicle_name: 'Innova', price: 1800000 },
  { vehicle_name: 'Tata Bus', price: 3500000 },
  { vehicle_name: 'Eicher Truck', price: 2800000 },
  { vehicle_name: 'Bolero Pickup', price: 900000 }
]
```

8.

vehicles> db.two_wheelers.find({manufacturer:"TVS"})

```
[
  {
    _id: ObjectId('69b19624072b2d5a197c2909'),
    bike_name: 'Apache RTR 160',
    model: 'gear',
    category: '150cc',
    colors_available: [ 'red', 'black' ],
    manufacturer: 'TVS',
    performance: 9,
    timestamp: ISODate('2021-03-20T00:00:00.000Z'),
    price: 120000
  },
  {
    _id: ObjectId('69b19624072b2d5a197c290a'),
    bike_name: 'Jupiter',
    model: 'gearless',
    category: '125cc',
    colors_available: [ 'blue', 'white' ],
    manufacturer: 'TVS',
    performance: 8,
    timestamp: ISODate('2018-07-11T00:00:00.000Z'),
    price: 78000
  }
]
```

9.

vehicles> db.four_wheelers.find({variants:"diesel"})

```
[
  {
    _id: ObjectId('69b1962d072b2d5a197c290d'),
    vehicle_name: 'Innova',
    model: 'own',
    category: 'car',
    variants: [ 'diesel', 'zxi' ],
    manufacturer: 'Toyota',
    performance: 9,
    timestamp: ISODate('2020-08-15T00:00:00.000Z'),
    price: 1800000
  },
  {
    _id: ObjectId('69b1962d072b2d5a197c290e'),
    vehicle_name: 'Tata Bus',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Tata',
    performance: 7,
    timestamp: ISODate('2018-09-01T00:00:00.000Z'),
    price: 3500000
  },
  {
    _id: ObjectId('69b1962d072b2d5a197c290f'),
    vehicle_name: 'Eicher Truck',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Eicher',
    performance: 8,
    timestamp: ISODate('2019-11-05T00:00:00.000Z'),
    price: 2800000
  },
  {
    _id: ObjectId('69b1962d072b2d5a197c2910'),
    vehicle_name: 'Bolero Pickup',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'diesel' ],
    manufacturer: 'Mahindra',
```

```
    performance: 8,
    timestamp: ISODate('2022-01-10T00:00:00.000Z'),
    price: 900000
  }
]
```

9. vehicles> db.two_wheelers.find(

| {performance:{$gt:5}},

| {bike_name:1, category:1, manufacturer:1, _id:0}

|)

|

| db.four_wheelers.find(

| {performance:{$gt:5}},

| {vehicle_name:1, category:1, manufacturer:1, _id:0}

|)

```
  { vehicle_name: 'Swift', category: 'car', manufacturer: 'Maruti' },
  { vehicle_name: 'Innova', category: 'car', manufacturer: 'Toyota' },
  { vehicle_name: 'Tata Bus', category: 'bus', manufacturer: 'Tata' },
  {
    vehicle_name: 'Eicher Truck',
    category: 'heavy truck',
    manufacturer: 'Eicher'
  },
  {
    vehicle_name: 'Bolero Pickup',
    category: 'mini truck',
    manufacturer: 'Mahindra'
```

2.

CODE:

1.
vehicles> use animal

```
switched to db animal
```

2. animal> show dbs

```
admin      40.00 KiB
config     92.00 KiB
local      40.00 KiB
vehicles   80.00 KiB
```

3. animal> db.createCollection("wild_animals", { capped: true, size: 50000,max: 100 })

|

| db.createCollection("domestic_animals")

{ ok: 1 }

animal> db.wild_animals.insertMany([

| {

| animal_name:"Lion",

| nature:"harm",

| favorite_foods:["meat","deer"],

| care_taker_name:"Ramesh",

| life_span:14,

| timestamp:new Date(),

| expenses:50000

| },

|

| {

| animal_name:"Tiger",

| nature:"harm",

| favorite_foods:["meat","deer"],

| care_taker_name:"Suresh",

```
| life_span:15,
| timestamp:new Date(),
| expenses:60000
| },
|
| {
| animal_name:"Elephant",
| nature:"harmless",
| favorite_foods:["grass","fruits"],
| care_taker_name:"Mahesh",
| life_span:60,
| timestamp:new Date(),
| expenses:70000
| },
|
| {
| animal_name:"Leopard",
| nature:"harm",
| favorite_foods:["meat"],
| care_taker_name:"Ramesh",
| life_span:12,
| timestamp:new Date(),
| expenses:45000
| },
|
| {
| animal_name:"Bear",
```

| nature:"harm",

| favorite_foods:["fish","honey"],

| care_taker_name:"Kiran",

| life_span:20,

| timestamp:new Date(),

| expenses:30000

| }

|])

```
acknowledged: true,
insertedIds: {
  '0': ObjectId('69b19a11072b2d5a197c2911'),
  '1': ObjectId('69b19a11072b2d5a197c2912'),
  '2': ObjectId('69b19a11072b2d5a197c2913'),
  '3': ObjectId('69b19a11072b2d5a197c2914'),
  '4': ObjectId('69b19a11072b2d5a197c2915')
}
```

4.animal> db.domestic_animals.insertMany([

| {

| animal_name:"Dog",

| gender:"male",

| favorite_foods:["meat","biscuits"],

| animal_petname:"Tommy",

| life_span:13,

| timestamp:new Date(),

| expenses:5000

| },

|

| {

| animal_name:"Cat",

| gender:"female",

| favorite_foods:["fish","milk"],

| animal_petname:"Kitty",

| life_span:15,

| timestamp:new Date(),

| expenses:3000

| },

|

| {

| animal_name:"Cow",

| gender:"female",

| favorite_foods:["grass"],

| animal_petname:"Gauri",

| life_span:20,

| timestamp:new Date(),

| expenses:8000

| },

|

| {

| animal_name:"Goat",

| gender:"male",

| favorite_foods:["leaves"],

| animal_petname:"Chotu",

| life_span:12,

| timestamp:new Date(),

| expenses:2000

| },

|

| {

| animal_name:"Horse",

| gender:"male",

| favorite_foods:["grass","grain"],

| animal_petname:"Raja",

| life_span:25,

| timestamp:new Date(),

| expenses:10000

| }

|])

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b19a16072b2d5a197c2916'),
    '1': ObjectId('69b19a16072b2d5a197c2917'),
    '2': ObjectId('69b19a16072b2d5a197c2918'),
    '3': ObjectId('69b19a16072b2d5a197c2919'),
    '4': ObjectId('69b19a16072b2d5a197c291a')
  }
}
```

6.

animal> db.wild_animals.find()

| db.domestic_animals.find()

```
[
  {
    _id: ObjectId('69b19a16072b2d5a197c2916'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'biscuits' ],
    animal_petname: 'Tommy',
    life_span: 13,
    timestamp: ISODate('2026-03-11T16:36:38.954Z'),
    expenses: 5000
  },
  {
    _id: ObjectId('69b19a16072b2d5a197c2917'),
    animal_name: 'Cat',
    gender: 'female',
    favorite_foods: [ 'fish', 'milk' ],
    animal_petname: 'Kitty',
    life_span: 15,
    timestamp: ISODate('2026-03-11T16:36:38.954Z'),
    expenses: 3000
  },
  {
    _id: ObjectId('69b19a16072b2d5a197c2918'),
    animal_name: 'Cow',
    gender: 'female',
    favorite_foods: [ 'grass' ],
    animal_petname: 'Gauri',
    life_span: 20,
    timestamp: ISODate('2026-03-11T16:36:38.954Z'),
    expenses: 8000
  }
```

```
},
{
  _id: ObjectId('69b19a16072b2d5a197c2919'),
  animal_name: 'Goat',
  gender: 'male',
  favorite_foods: [ 'leaves' ],
  animal_petname: 'Chotu',
  life_span: 12,
  timestamp: ISODate('2026-03-11T16:36:38.954Z'),
  expenses: 2000
},
{
  _id: ObjectId('69b19a16072b2d5a197c291a'),
  animal_name: 'Horse',
  gender: 'male',
  favorite_foods: [ 'grass', 'grain' ],
  animal_petname: 'Raja',
  life_span: 25,
  timestamp: ISODate('2026-03-11T16:36:38.954Z'),
  expenses: 10000
}
```

7.

animal> db.wild_animals.find({}, {animal_name:1, expenses:1, _id:0})

| db.domestic_animals.find({}, {animal_name:1, expenses:1, _id:0})

```
[
  { animal_name: 'Dog', expenses: 5000 },
  { animal_name: 'Cat', expenses: 3000 },
  { animal_name: 'Cow', expenses: 8000 },
  { animal_name: 'Goat', expenses: 2000 },
  { animal_name: 'Horse', expenses: 10000 }
]
```

8.

animal> db.domestic_animals.find({life_span:15})

```
[
  {
    _id: ObjectId('69b19a16072b2d5a197c2917'),
    animal_name: 'Cat',
    gender: 'female',
    favorite_foods: [ 'fish', 'milk' ],
    animal_petname: 'Kitty',
    life_span: 15,
    timestamp: ISODate('2026-03-11T16:36:38.954Z'),
    expenses: 3000
  }
]
```

9.

animal> db.wild_animals.find({care_taker_name:"Ramesh"})

```
[
  {
    _id: ObjectId('69b19a11072b2d5a197c2911'),
    animal_name: 'Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer' ],
    care_taker_name: 'Ramesh',
    life_span: 14,
    timestamp: ISODate('2026-03-11T16:36:33.189Z'),
    expenses: 50000
  },
  {
    _id: ObjectId('69b19a11072b2d5a197c2914'),
    animal_name: 'Leopard',
    nature: 'harm',
    favorite_foods: [ 'meat' ],
    care_taker_name: 'Ramesh',
    life_span: 12,
    timestamp: ISODate('2026-03-11T16:36:33.189Z'),
    expenses: 45000
  }
]
```

10.

animal> db.wild_animals.find(

| {life_span:{$gt:5}},

| {animal_name:1, favorite_foods:1, expenses:1, _id:0}

|)

|

| db.domestic_animals.find(

| {life_span:{$gt:5}},

| {animal_name:1, favorite_foods:1, expenses:1, _id:0}

|)

```
[
  {
    animal_name: 'Dog',
    favorite_foods: [ 'meat', 'biscuits' ],
    expenses: 5000
  },
  {
    animal_name: 'Cat',
    favorite_foods: [ 'fish', 'milk' ],
    expenses: 3000
  },
  { animal_name: 'Cow', favorite_foods: [ 'grass' ], expenses: 8000 },
  { animal_name: 'Goat', favorite_foods: [ 'leaves' ], expenses: 2000 },
  {
    animal_name: 'Horse',
    favorite_foods: [ 'grass', 'grain' ],
    expenses: 10000
  }
]
```

THE END